

Rezolvare — Model de antrenament SAE

1. $G(s) = \frac{1}{s(s+5)}, T_e = 0.1s$

a) Funcția de transfer în circuit închis (metoda trapezului)

$$s = \frac{2}{T_e} \cdot \frac{z-1}{z+1} = 20 \frac{z-1}{z+1}$$

$$s+5 = \frac{20(z-1)+5(z+1)}{z+1} = \frac{25z-15}{z+1} = \frac{5(5z-3)}{z+1}$$

$$G(z) = \frac{1}{s(s+5)} = \frac{(z+1)^2}{20(z-1) \cdot 5(5z-3)} = \frac{(z+1)^2}{100(z-1)(5z-3)} = \frac{z^2+2z+1}{500z^2-800z+300}$$

$$H(z) = \frac{G(z)}{1+G(z)} = \frac{z^2+2z+1}{(500z^2-800z+300)+(z^2+2z+1)}$$

$$H(z) = \frac{(z+1)^2}{501z^2-798z+301}$$

b) Eroare staționară

$$K_p = \lim_{z \rightarrow 1} G(z) = \infty \quad K_v = \frac{1}{T_e} \lim_{z \rightarrow 1} (z-1)G(z) = \frac{1}{T_e} \cdot \frac{(z+1)^2}{100(5z-3)} \Big|_{z=1} = \frac{1}{0.1} \cdot \frac{4}{200} = 0.2$$

$$K_a = \frac{1}{T_e^2} \lim_{z \rightarrow 1} (z-1)^2 G(z) = 0$$

$$e_{ss}(\text{treaptă}) = \frac{1}{1+K_p} = 0 \quad e_{ss}(\text{rampă}) = \frac{1}{K_v} = \frac{1}{0.2} = 5 \quad e_{ss}(\text{parabolă}) = \frac{1}{K_a} = \infty$$

c) Răspunsul la impuls unitar Dirac

$$R(z) = 1 \Rightarrow Y(z) = H(z)$$

$$z_{1,2} = \frac{798 \pm \sqrt{798^2 - 4 \cdot 501 \cdot 301}}{1002} = \frac{798 \pm 40\sqrt{21}}{1002}$$

$$z_1 \approx 0.9793 \quad z_2 \approx 0.6135 \quad (\text{sistem stabil})$$

$$\frac{Y(z)}{z} = \frac{z^2+2z+1}{z(501z^2-798z+301)} = \frac{A}{z} + \frac{B}{z-z_1} + \frac{C}{z-z_2}$$

$$A = \frac{N(0)}{D(0)} = \frac{1}{301} \approx 0.0033, \quad B = \frac{N(z_1)}{z_1 \cdot 501(z_1 - z_2)} \approx 0.0218, \quad C = \frac{N(z_2)}{z_2 \cdot 501(z_2 - z_1)} \approx -0.0232$$

$$N(z) = (z+1)^2, \quad D(z) = 501z^2 - 798z + 301$$

$$Y(z) = A + \frac{Bz}{z-z_1} + \frac{Cz}{z-z_2}$$

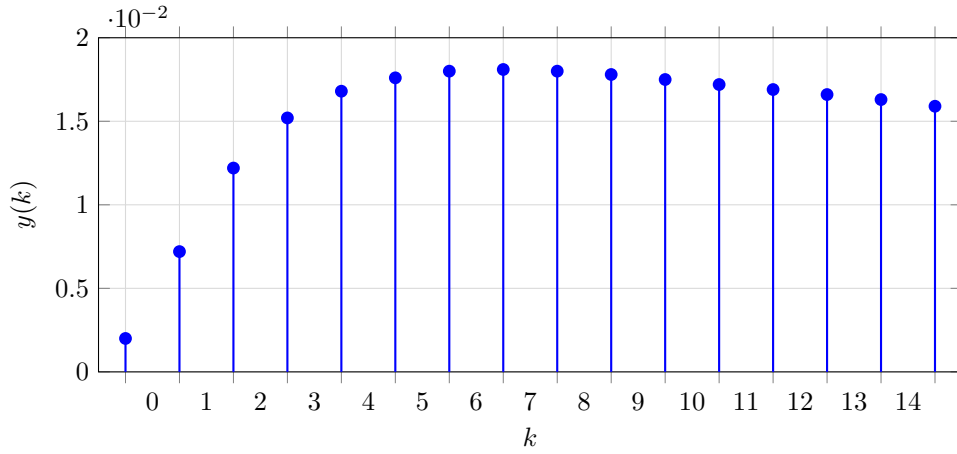
$$Y(z) = 0.0033 + \frac{0.0218z}{z-0.9793} - \frac{0.0232z}{z-0.6135}$$

$$\mathcal{Z}^{-1}\{1\} = \delta(k), \quad \mathcal{Z}^{-1}\{z/(z-a)\} = a^k u(k)$$

$$y(k) = 0.0033 \delta(k) + 0.0218 (0.9793)^k - 0.0232 (0.6135)^k, \quad k \geq 0$$

Verificare: $\sum_k y(k) = H(1) = \frac{4}{501-798+301} = \frac{4}{4} = 1$, câștig static unitar — consistent cu $e_{ss}(\text{treaptă}) = 0$.

k	0	1	2	3	4	5	6	7
$y(k)$	0.0020	0.0072	0.0122	0.0152	0.0168	0.0177	0.0180	0.0181
k	8	9	10	11	12	13	14	15
$y(k)$	0.0180	0.0178	0.0175	0.0172	0.0169	0.0166	0.0163	0.0159



2. $G_d(z) = \frac{0.2z + 0.07}{z^2 - 0.5z + 0.05}$, $T_e = 1s$ (circuit deschis, reacție unitară)

$$z = \frac{1 + \frac{T_e}{2}w}{1 - \frac{T_e}{2}w} \quad T_e=1 \quad \frac{1 + 0.5w}{1 - 0.5w} = \frac{N}{D}, \quad N = 1 + 0.5w, \quad D = 1 - 0.5w$$

$$0.2z + 0.07 = \frac{0.2N + 0.07D}{D} = \frac{0.27 + 0.07w}{D}$$

$$z^2 - 0.5z + 0.05 = \frac{N^2 - 0.5ND + 0.05D^2}{D^2} = \frac{0.39w^2 + 0.95w + 0.55}{D^2}$$

$$G_d(w) = \frac{(0.27 + 0.07w) D}{0.39w^2 + 0.95w + 0.55} = \frac{-0.03w^2 - 0.07w + 0.27}{0.39w^2 + 0.95w + 0.55}$$

$$N(w) = 0 \Rightarrow w_{z1} = 2, \quad w_{z2} = -4.15 \quad D(w) = 0 \Rightarrow w_{p1} = -0.94, \quad w_{p2} = -1.51$$

$$G_d(w) = \underbrace{\frac{0.27}{0.55}}_{K_B=0.49} \cdot \frac{\left(1 - \frac{w}{2}\right) \left(1 + \frac{w}{4.15}\right)}{\left(1 + \frac{w}{0.94}\right) \left(1 + \frac{w}{1.51}\right)}$$

$w_{z1} = 2 > 0 \Rightarrow$ zerou neminim-fazic. Frecvențe de frângere (rad/s): $\omega_{p1} = 0.94$, $\omega_{p2} = 1.51$, $\omega_{z1} = 2$, $\omega_{z2} = 4.15$.

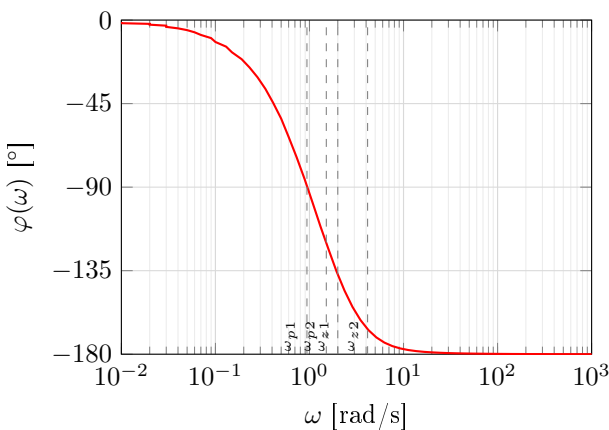
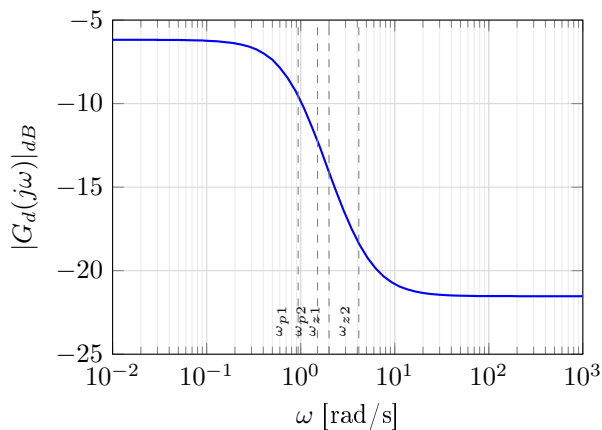
Panta asimptotică a modului (nivel de start $20 \lg K_B = -6.18\text{dB}$):

$$\underbrace{0}_{\omega < 0.94} \rightarrow \underbrace{-20}_{0.94 < \omega < 1.51} \rightarrow \underbrace{-40}_{1.51 < \omega < 2} \rightarrow \underbrace{-20}_{2 < \omega < 4.15} \rightarrow \underbrace{0}_{\omega > 4.15} \text{ dB/dec}$$

Nivel final ($\omega \rightarrow \infty$): $20 \lg \left| \frac{-0.03}{0.39} \right| = -21.53\text{dB}$ (raport coeficienți dominanți). Fază asimptotică ($\omega \rightarrow \infty$):

2 poli(-90°) + 1 zerou m.f.($+90^\circ$) + 1 zerou n.m.f.(-90°) = -180° .

ω	0.01	0.1	0.5	1	2	5	10	50	100
$ G_d _{dB}$	-6.18	-6.24	-7.39	-9.84	-14.09	-19.14	-20.80	-21.50	-21.52
$\varphi [^\circ]$	-1.1	-11.4	-53.5	-93.3	-137.0	-170.5	-177.3	-179.7	-179.8



$$3. G_d(z) = \frac{k(0.13z + 0.02)}{z^2 - 0.39z + 0.007}, T_e = 1s$$

$$\text{Poli: } z^2 - 0.39z + 0.007 = 0 \Rightarrow p_{1,2} = \frac{0.39 \pm \sqrt{0.39^2 - 4 \cdot 0.007}}{2}$$

$$p_1 = 0.37 \quad p_2 = 0.02 \quad z_0 = -\frac{0.02}{0.13} = -0.15 \text{ (zero)}$$

a) Numărul de ramuri

$$n = 2, m = 1 \Rightarrow \text{nr. ramuri} = \max(n, m) = 2.$$

Ramuri din $p_1 = 0.37, p_2 = 0.02$; se termină în $z_0 = -0.15$ și în $z \rightarrow \infty$.

b) Porțiunea de pe axa reală

$$(p_2, p_1) = (0.02, 0.37) \quad \text{și} \quad (-\infty, z_0) = (-\infty, -0.15)$$

c) Asimptote

$$n - m = 1, \quad \theta = \frac{180^\circ(2k+1)}{n-m} = 180^\circ$$

$$\sigma_a = \frac{\sum p_i - \sum z_i}{n-m} = \frac{(0.37 + 0.02) - (-0.15)}{1} = 0.54$$

1 asimptotă, 180° , $\sigma_a = 0.54$.

d) Punctele de ramificație

$$k(z) = -\frac{N(z)}{D(z)}, \quad \frac{dk}{dz} = 0$$

$$N(z) = z^2 - 0.39z + 0.007, \quad D(z) = 0.13z + 0.02$$

$$0.13z^2 + 0.04z - 0.01 = 0$$

$$z_{b1} = 0.15 \quad (k \approx 0.73) \quad z_{b2} = -0.45 \quad (k \approx 10.0)$$

e) Verificare cerc & trasare

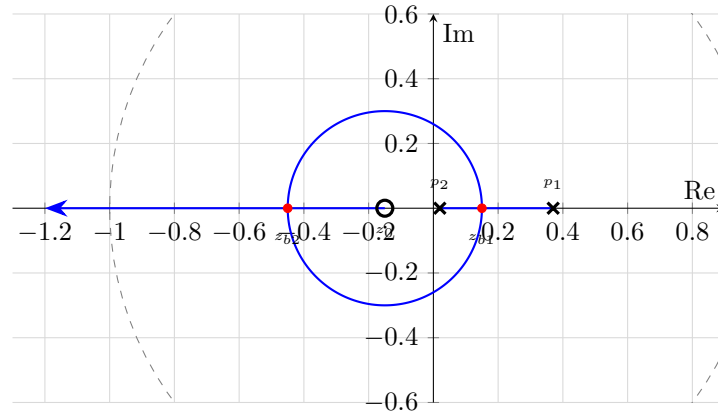
$$z = z_0 + w: \quad w^2 + w[2z_0 - p_1 - p_2 + k] + (z_0 - p_1)(z_0 - p_2) = 0$$

$$|w|^2 = w\bar{w} = (z_0 - p_1)(z_0 - p_2) \quad (\text{independent de } k)$$

$$r = \sqrt{(z_0 - p_1)(z_0 - p_2)} = \sqrt{(-0.15 - 0.37)(-0.15 - 0.02)}$$

$$r = \sqrt{0.52 \cdot 0.17} = 0.30$$

Verificare: $|z_{b1} - z_0| = 0.30$ și $|z_{b2} - z_0| = 0.30$ — exact raza.



(cerc punctat = cerc unitate.)

$$4. G_0(z) = \frac{0.02z + 0.02}{z^2 - 1.49z + 0.54}$$

a) Model la stare, forma complet controlabilă

$$G_0(z) = \frac{Y(z)}{U(z)} = \frac{b_1 z + b_0}{z^2 + a_1 z + a_0}, \quad a_1 = -1.49, \quad a_0 = 0.54, \quad b_1 = 0.02, \quad b_0 = 0.02$$

$$w(k+2) - a_1 w(k+1) + a_0 w(k) = u(k), \quad y(k) = b_1 w(k+1) + b_0 w(k)$$

$$w(k+2) = 1.49 w(k+1) - 0.54 w(k) + u(k)$$

$$x_1(k) = w(k), \quad x_2(k) = w(k+1)$$

$$x_1(k+1) = x_2(k) \quad x_2(k+1) = -0.54 x_1(k) + 1.49 x_2(k) + u(k) \quad y(k) = 0.02 x_1(k) + 0.02 x_2(k)$$

$$A = \begin{pmatrix} 0 & 1 \\ -0.54 & 1.49 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad C = (0.02 \quad 0.02), \quad D = 0$$

$$AB = \begin{pmatrix} 1 \\ 1.49 \end{pmatrix}, \quad Q_c = (B \quad AB) = \begin{pmatrix} 0 & 1 \\ 1 & 1.49 \end{pmatrix}, \quad \det Q_c = -1 \neq 0 \Rightarrow \text{complet controlabil}$$

b) Observabilitate

$$CA = (0.02 \quad 0.02) \begin{pmatrix} 0 & 1 \\ -0.54 & 1.49 \end{pmatrix} = (-0.0108 \quad 0.0498)$$

$$Q_o = \begin{pmatrix} C \\ CA \end{pmatrix} = \begin{pmatrix} 0.02 & 0.02 \\ -0.0108 & 0.0498 \end{pmatrix}, \quad \det Q_o = 0.02 \cdot 0.0498 - 0.02 \cdot (-0.0108) = 0.0012 \neq 0$$

Sistemul este **complet observabil**.

$$5. G(z) = \frac{z^2 - 1.39z + 0.52}{z^2 - 1.43z + 0.49}$$

a) Model la stare, forma complet observabilă

$$G(z) = \frac{Y(z)}{U(z)} = \frac{b_2 z^2 + b_1 z + b_0}{z^2 + a_1 z + a_0}, \quad a_1 = -1.43, \quad a_0 = 0.49, \quad b_2 = 1, \quad b_1 = -1.39, \quad b_0 = 0.52$$

$$B(z) - b_n A(z) = (z^2 - 1.39z + 0.52) - (z^2 - 1.43z + 0.49) = 0.04z + 0.03 = B'(z)$$

$$D = b_n = 1, \quad b'_1 = b_1 - b_n a_1 = -1.39 - (-1.43) = 0.04, \quad b'_0 = b_0 - b_n a_0 = 0.52 - 0.49 = 0.03$$

$$G(z) = D + \frac{B'(z)}{A(z)} = 1 + \frac{0.04z + 0.03}{z^2 - 1.43z + 0.49}$$

$$x_1(k+1) = -a_0 x_2(k) + b'_0 u(k) \quad x_2(k+1) = x_1(k) - a_1 x_2(k) + b'_1 u(k) \quad y_{sp}(k) = x_2(k)$$

$$y(k) = x_2(k) + Du(k)$$

$$A_o = \begin{pmatrix} 0 & -a_0 \\ 1 & -a_1 \end{pmatrix} = \begin{pmatrix} 0 & -0.49 \\ 1 & 1.43 \end{pmatrix}, \quad B_o = \begin{pmatrix} b'_0 \\ b'_1 \end{pmatrix} = \begin{pmatrix} 0.03 \\ 0.04 \end{pmatrix}, \quad C_o = (0 \quad 1), \quad D = 1$$

Verificare: $(zI - A_o) = \begin{pmatrix} z & 0.49 \\ -1 & z - 1.43 \end{pmatrix}$, $\det = z^2 - 1.43z + 0.49 = A(z) \checkmark$.

$$C_o(zI - A_o)^{-1}B_o = \frac{[0 \quad 1] \begin{pmatrix} z - 1.43 & -0.49 \\ 1 & z \end{pmatrix} \begin{pmatrix} 0.03 \\ 0.04 \end{pmatrix}}{z^2 - 1.43z + 0.49} = \frac{0.03 + 0.04z}{z^2 - 1.43z + 0.49}$$

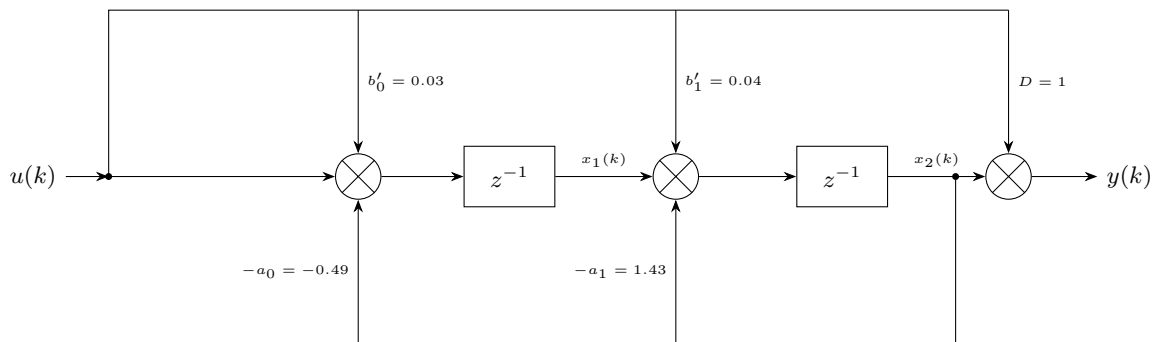
$$G(z) = C_o(zI - A_o)^{-1}B_o + D = 1 + \frac{0.04z + 0.03}{z^2 - 1.43z + 0.49} = \frac{z^2 - 1.39z + 0.52}{z^2 - 1.43z + 0.49} \quad \checkmark$$

$$C_o A_o = (0 \quad 1) \begin{pmatrix} 0 & -0.49 \\ 1 & 1.43 \end{pmatrix} = (1 \quad 1.43)$$

$$Q_o = \begin{pmatrix} C_o \\ C_o A_o \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1.43 \end{pmatrix}, \quad \det Q_o = 0 \cdot 1.43 - 1 \cdot 1 = -1 \neq 0 \Rightarrow \text{complet observabil}$$

b) Schema bloc

$$x_1(k+1) = -0.49x_2(k) + 0.03u(k) \quad x_2(k+1) = x_1(k) + 1.43x_2(k) + 0.04u(k) \quad y(k) = x_2(k) + u(k)$$



(formă canonică observabilă — Direct Form II transpusă.)

Teorie (pe scurt)

1. Transformarea planului s în planul z

$$x^*(t) = \sum_k x(kT_e)\delta(t - kT_e), \quad X^*(s) = \sum_k x(kT_e)e^{-skT_e} \quad (\text{periodică, } j\omega_e = j2\pi/T_e)$$

$$z = e^{sT_e} : \quad \text{Re}(s) = 0 \rightarrow |z| = 1, \quad \text{Re}(s) < 0 \rightarrow |z| < 1$$

2. Schema bloc a unui sistem de reglare numeric

$$R(z) \rightarrow \text{Sumator}(\pm) \rightarrow H_R(z) \rightarrow N/A + \text{ZOH} \rightarrow G(s) \rightarrow y(t) \rightarrow A/N \rightarrow (\text{reacție})$$

3. Etapele conversiei analog-numerice

$$\text{Eșantionare} \rightarrow \text{Reținere (S\&H)} \rightarrow \text{Cuantizare} \rightarrow \text{Codificare}$$

4. Elementele specifice unui sistem numeric

A/N (ADC), regulator numeric, N/A (DAC) + ERO/ZOH, eșantionator (perioadă T_e).

5. Eșantionorul ideal în schemă bloc

$$x^*(t) = x(t) \sum_k \delta(t - kT_e)$$